

```
1 A) if (grade >= 90) {  
    System.out.println("great job!");  
}
```

```
B) if (number < 20 || number > 50) {  
    System.out.println("error");  
}
```

```
C) if (y < 100) {  
    y += 2; // or y = y + 2;  
}
```

```
2. if (num1 > num2) {  
    System.out.println("first number is larger.");  
} else if (num2 > num1) {  
    System.out.println("second number is larger.");  
} else {  
    System.out.println("numbers are equal.");  
}
```

```
3.  
    A) if (num % 2 == 0) {  
        System.out.println("even number");  
    } else {  
        System.out.println("odd number");  
    }
```

```
    B) switch (expression) {  
        case value1:  
            // code to execute if expression matches value1  
            break;  
        case value2:  
            // code to execute if expression matches value2  
            break;  
        default:  
            // code to execute if none of the cases match  
            Break;
```

```
4.  
    A) int num1 = (int) (Math.random() * 50) + 1;  
    B) int num2 = (int) (Math.random() * 20) + 1;  
    C) int num3 = (int) (Math.random() * 100) + 1;  
    D) int num4 = (int) (Math.random() * ((20 - 10) + 1)) + 10;
```

```
5. Corrected code:  
if (age < 18) {
```

```
    System.out.println("child");  
} else if (age >= 18 && age < 65) {  
    System.out.println("Adult");  
} else {  
    System.out.println("senior");  
}
```

6.

- a) T
- b) F
- C) F
- D) T
- E) F
- F) T
- G) T